

Digital Image Processing:

3 Image Enhancement---Spatial domain[1]

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LABS!

Day: Thursday

Time: 8:00 – 9:40am

Room: Shiyan building #303

Matlab Image Processing Toolbox

Contents

Over the next few lectures we will look at image enhancement techniques working in the spatial domain:

1. What is **image enhancement**?
2. **Intensity transformation** enhancement
3. **Histogram** processing
4. **Neighbourhood operations(next class)**

Recall: Grey Levels

In our class, image grey level values we have said they are in the range $[0, 255]$

- Where 0 is black and 255 is white.

There is no reason why we have to use this range

- The range $[0, 255]$ stems from display technologies

For many of the image processing operations in this lecture grey levels are assumed to be given in the range $[0.0, 1.0]$

1 What is Image Enhancement?

Image enhancement: the process of making images more suitable for a specific application.

The reasons for doing this include:

- Highlighting interesting detail in images
- Removing noise from images
- Making images more visually appealing

Spatial & Frequency Domains

There are two broad categories of image enhancement techniques

(a) Spatial domain techniques

- Direct manipulation of image pixels: Image Plane

(b) Frequency domain techniques

- Manipulation of Fourier transform or wavelet transform of an image

This lecture we will concentrate on techniques that operate in the spatial domain.

Image enhancement

- Spatial domain methods

- (1) Point-based methods - intensity transformations

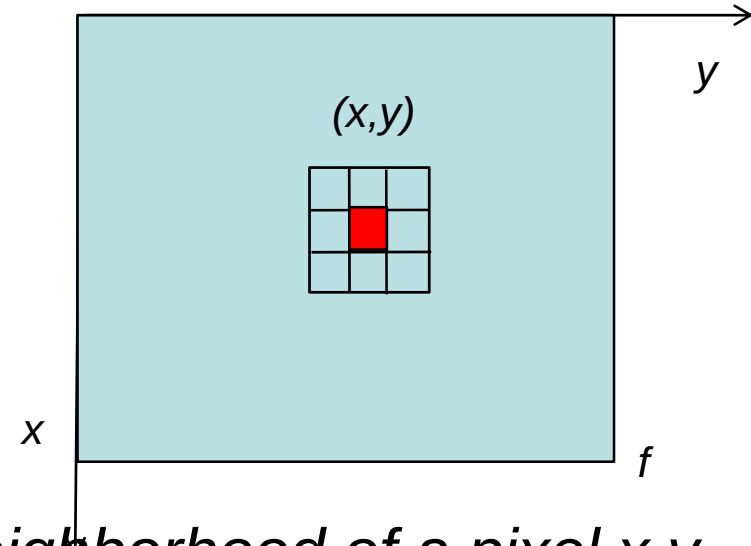
- Negative
 - Log transformation
 - Power-law
 - Contrast stretching
 - Gray-level slicing
 - Histogram equalization
 - Averaging

- (2) Mask-based (neighborhood-based) methods - spatial filter

- Frequency domain methods

Basics of spatial domain processing

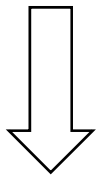
- $g(x,y) = T[f(x,y)]$
- $f(x,y)$: Input image
- $g(x,y)$: Output image
- T : operator on f defined over a neighborhood of a pixel x,y
 - ✓ standard approach: T is applied to a sub-image centred at (x,y) .
 - ✓ sub-image is called mask(kernel, filter, template, window)
 - ✓ mask processing or filtering
 - ✓ T can operate on a set of images



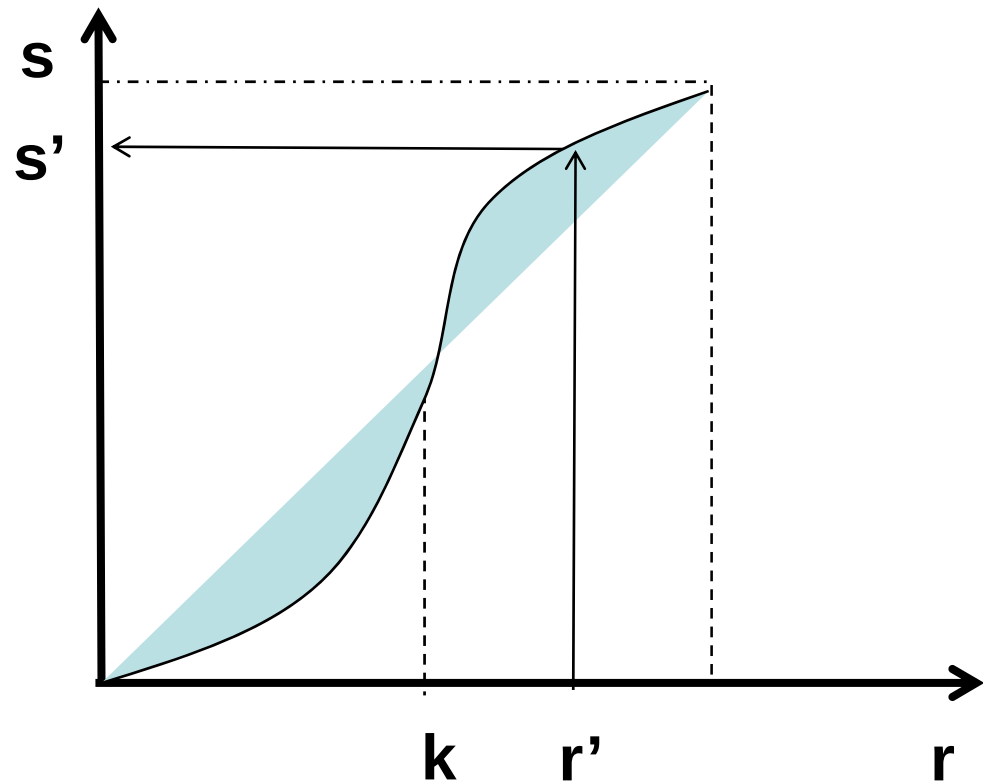
2 Intensity transformations

- Point processing: The simplest kind of range transformations are these independent of position x,y :

$$g(x,y) = T[f(x,y)]$$

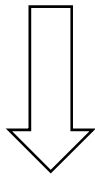


$$s = T(r)$$



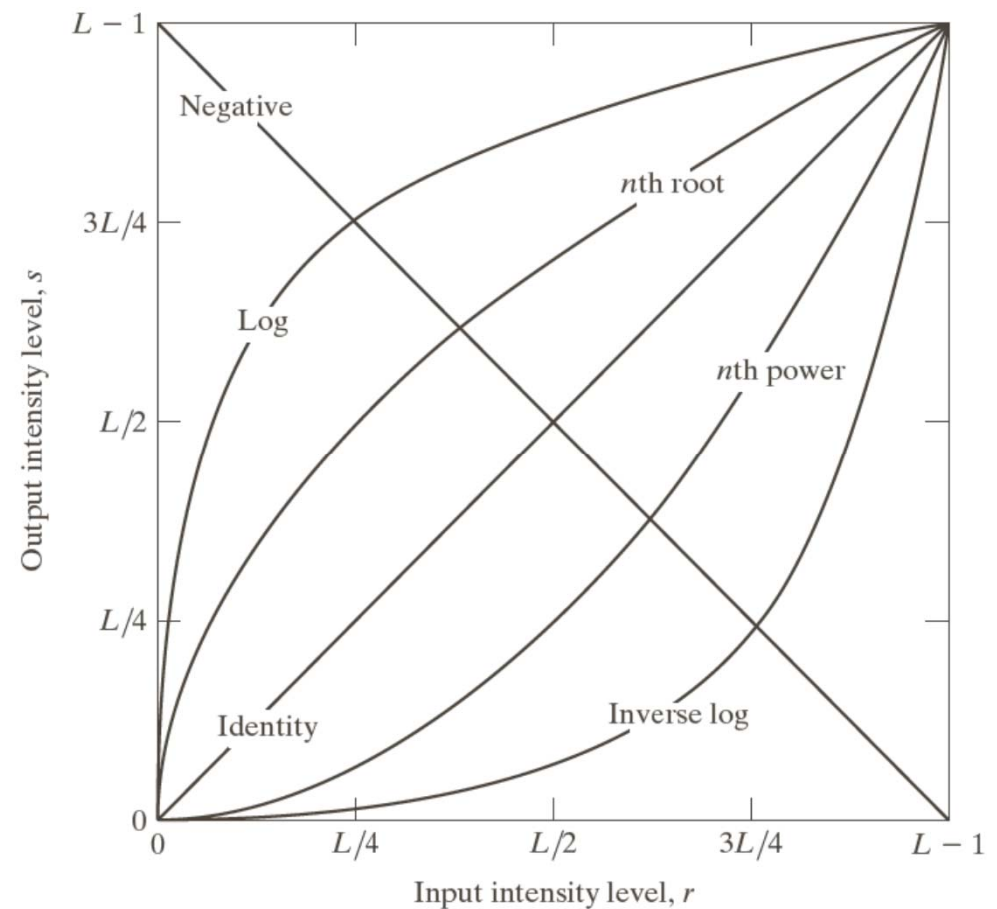
2 Intensity transformations

$$g(x,y) = T[f(x,y)]$$



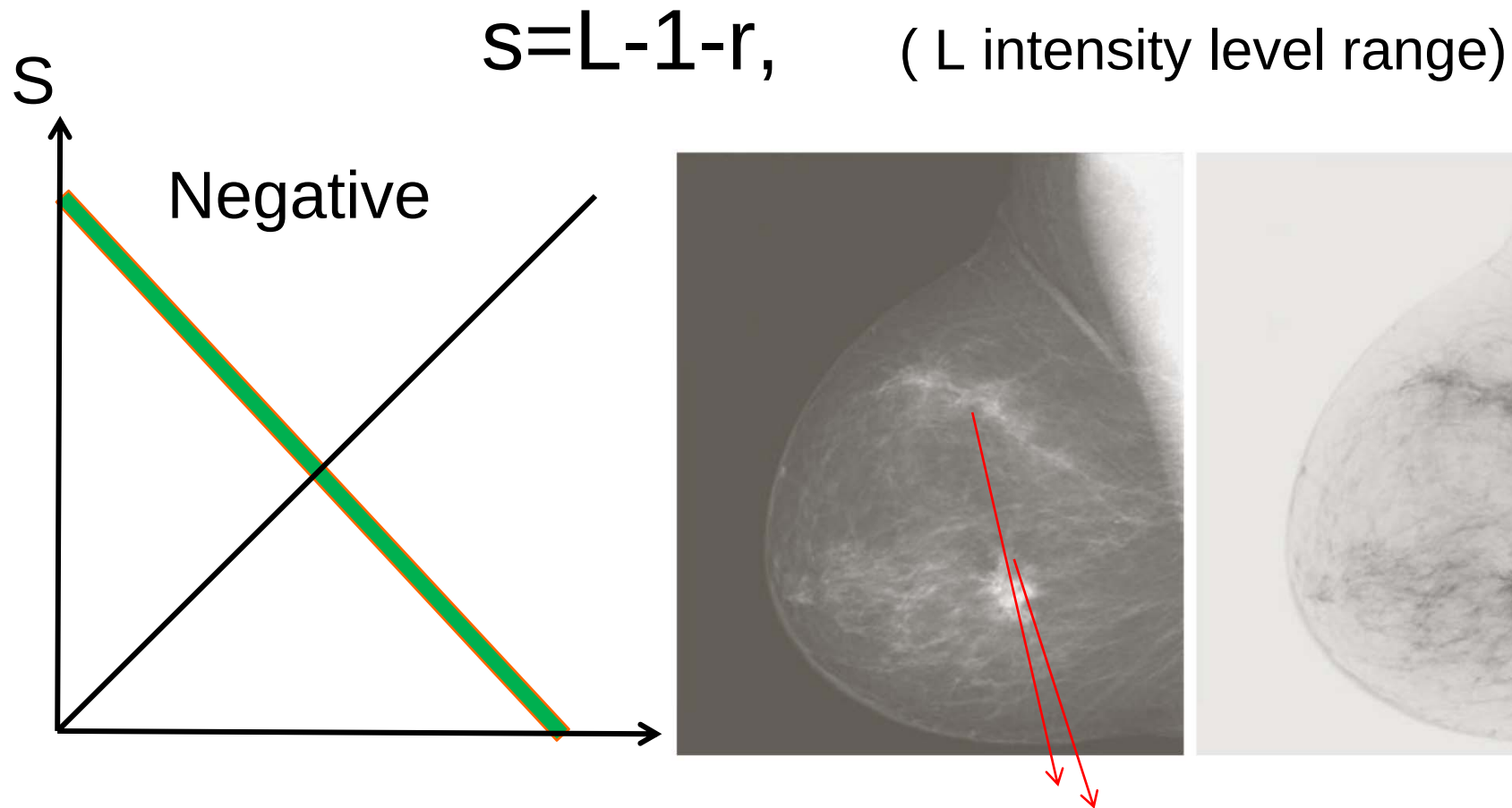
$$s = T(r)$$

FIGURE 3.3 Some basic intensity transformation functions. All curves were scaled to fit in the range shown.



2 Intensity transformations

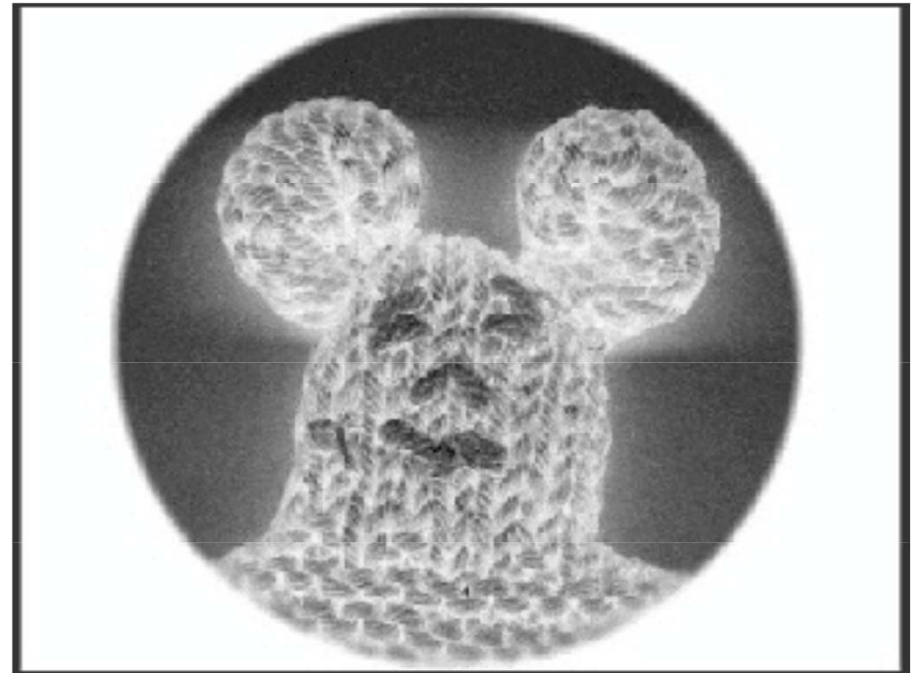
(1) Image Negatives



Interest area

2 Intensity transformations

(1) Image Negatives



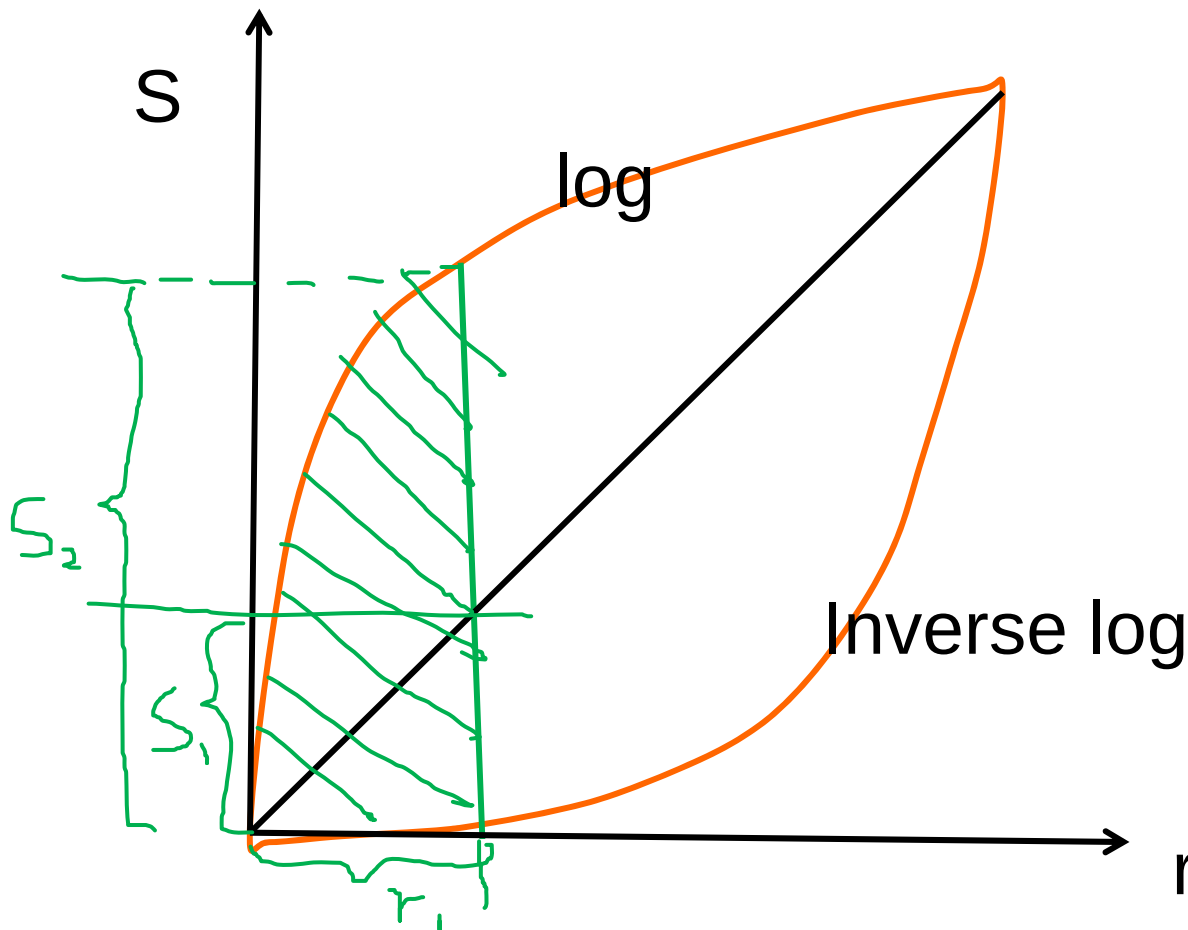
Suited for:

Enhancing white or gray detail embedded in dark regions of an image with an black areas dominant in size.

2 Intensity transformations

(2) Log Transformations

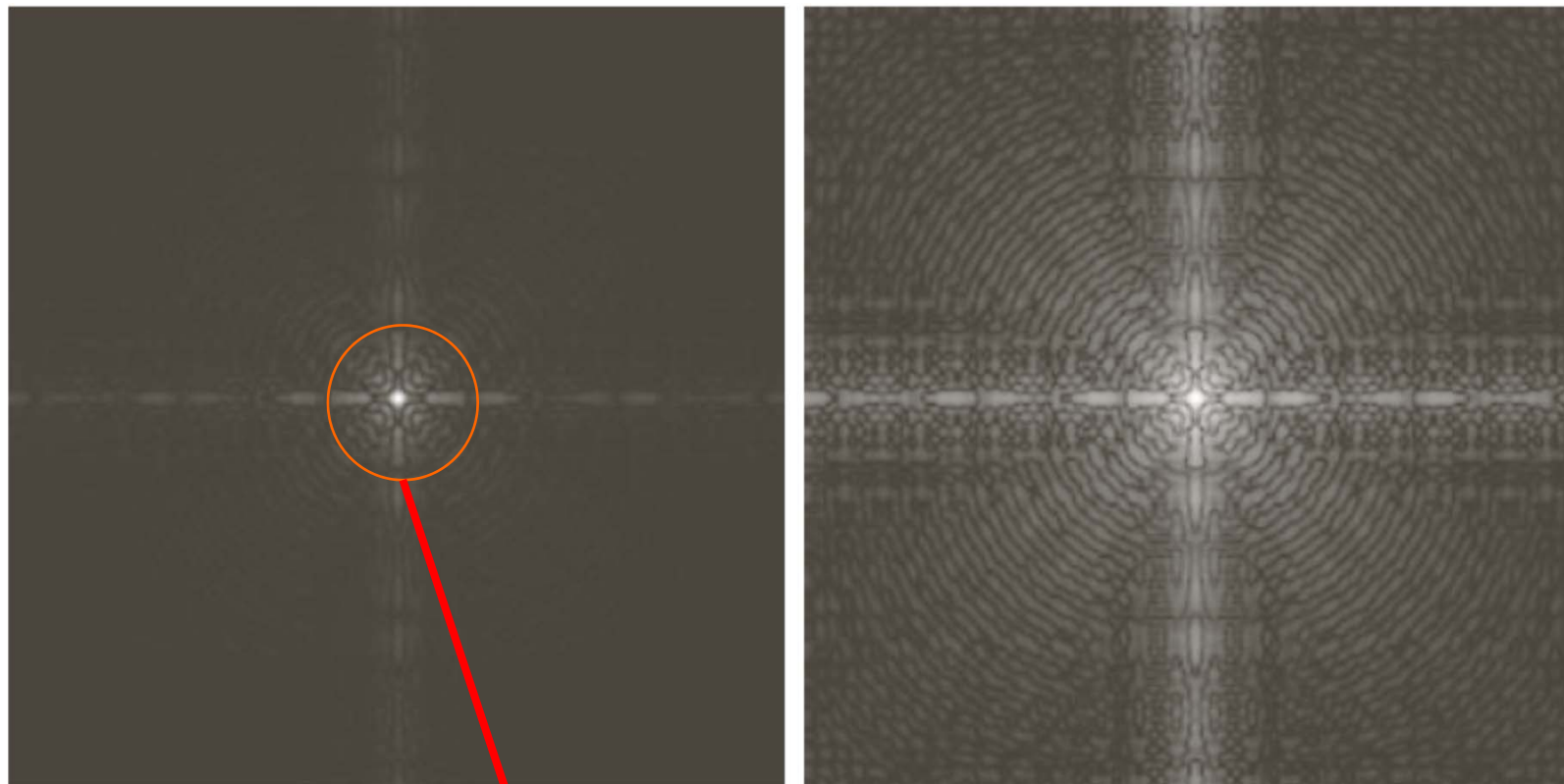
$$s = c \log(1+r), \quad (c \text{ is constant})$$



2 Intensity transformations

(2) Log Transformations

$$s = c \log(1 + r), \quad (c \text{ is constant})$$



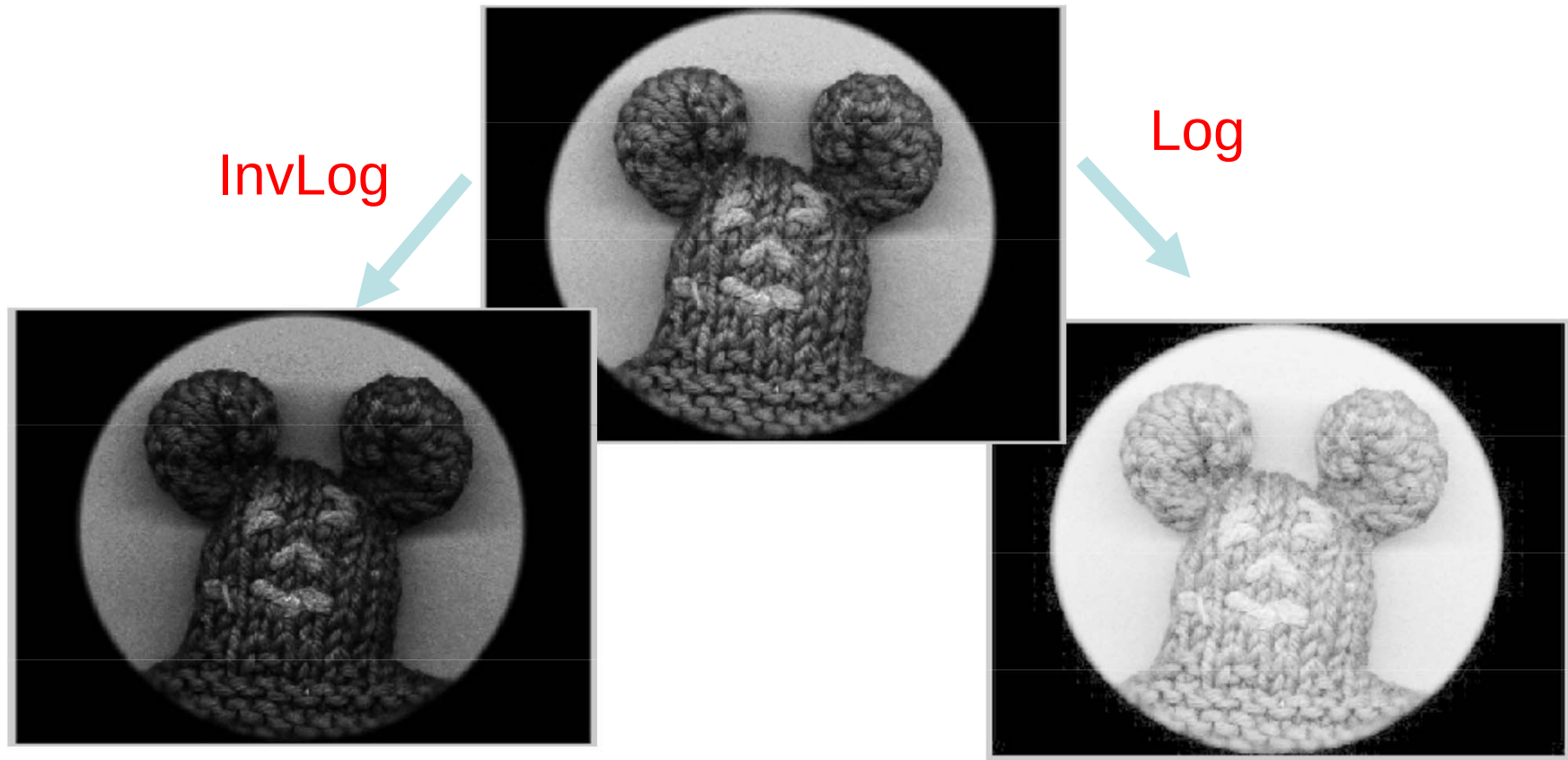
a b

FIGURE 3.5
(a) Fourier spectrum.
(b) Result of applying the log transformation in Eq. (3.2-2) with $c = 1$.

Interest area

2 Intensity transformations

(2) Log Transformations



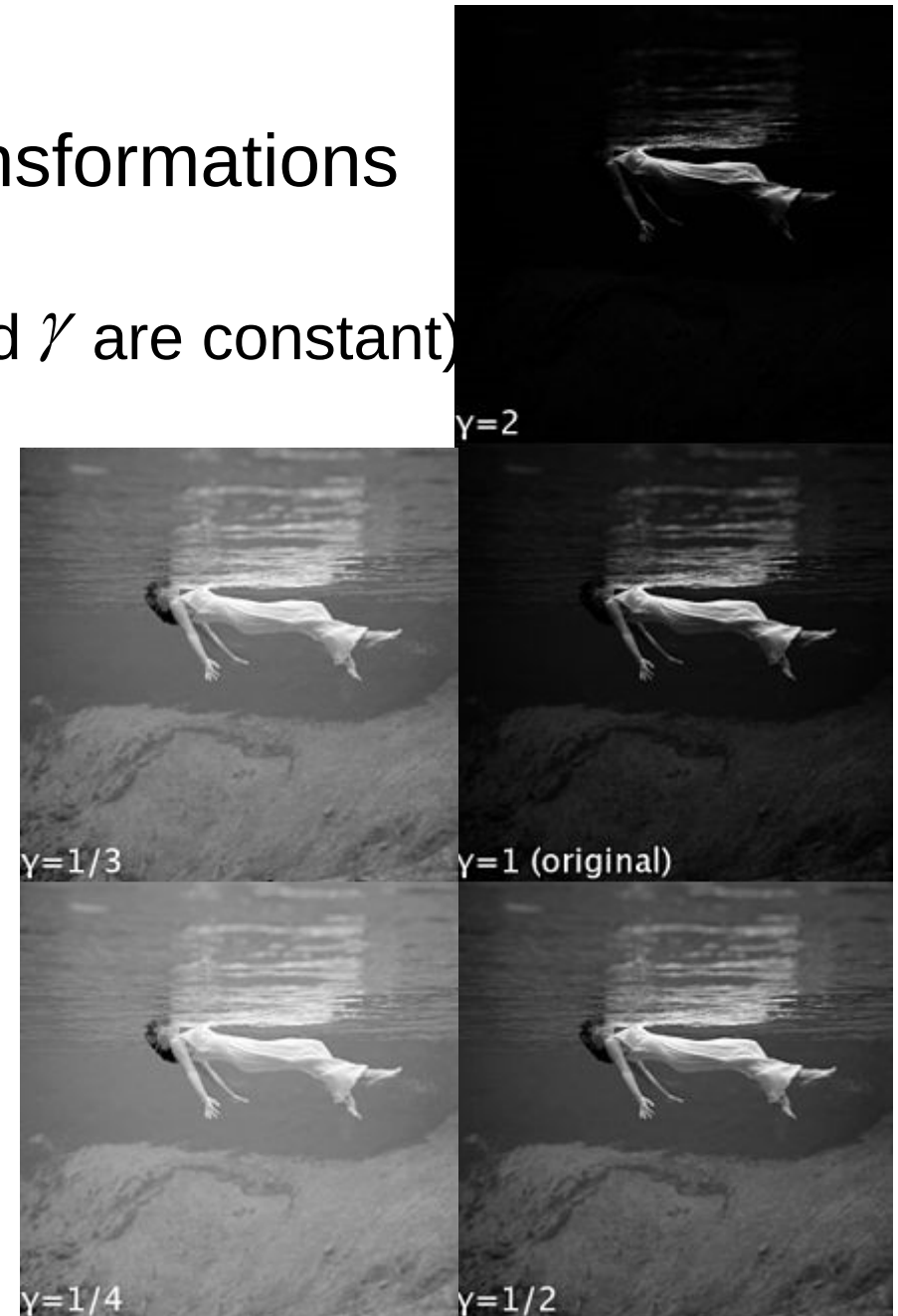
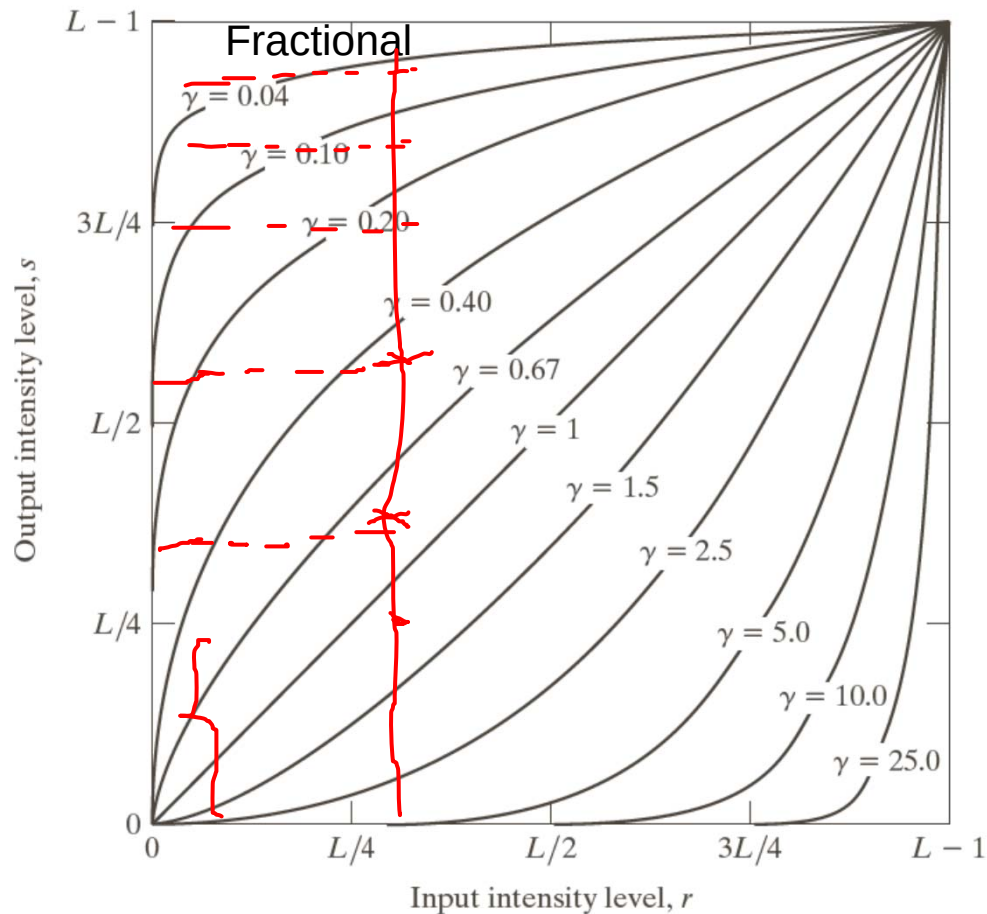
Suited for:

- a. Expand values of dark pixels.
- b. Compresses the dynamic range.

2 Intensity transformations

(3) Power-law (Gamma) Transformations

$$S = c r^{\gamma} \quad (c \text{ and } \gamma \text{ are constant})$$



Why Gamma Correction?

Devices for image capture, printing and display respond according to a power law.

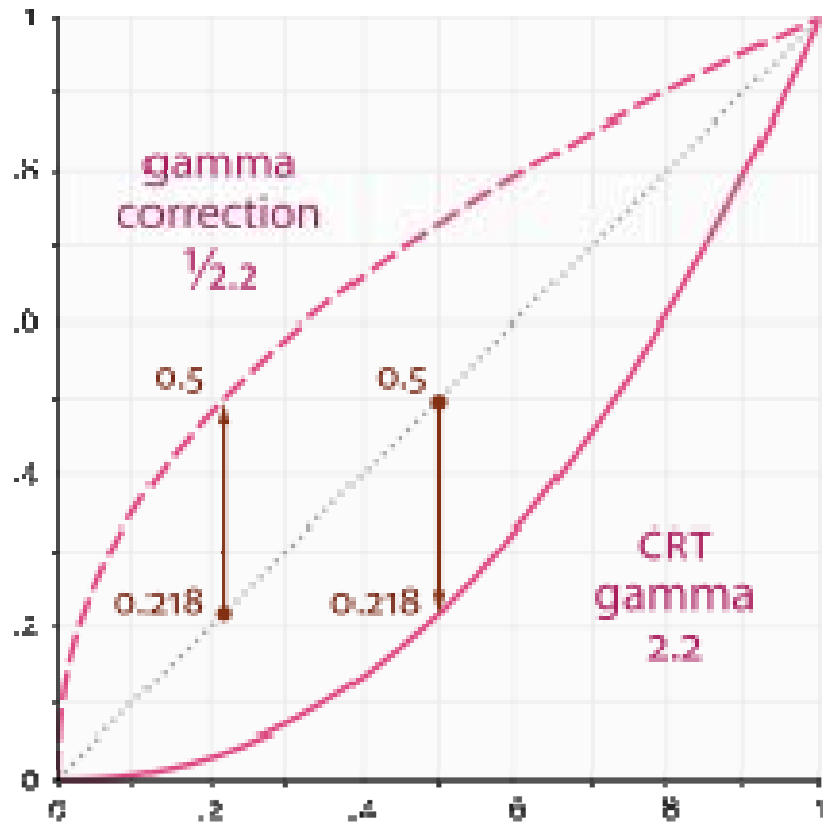


Why Gamma Correction?

— CRT monitor

Intensity-to-voltage response:

a power function with exponent varying from 1.8~2.5



↓
CRT output is
darker or brighter?

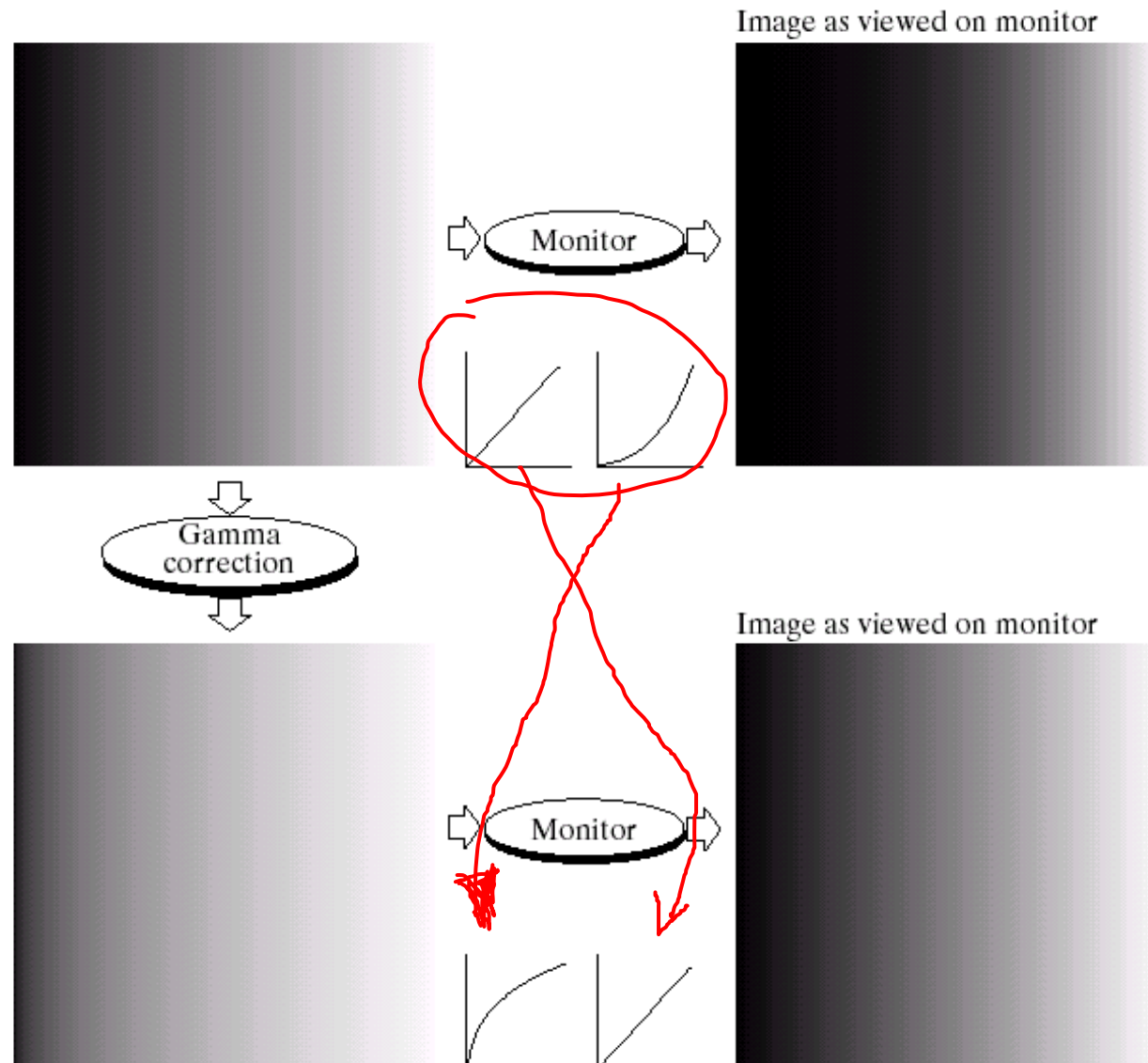
Why Gamma Correction?

— CRT monitor

a b
c d

FIGURE 3.7

(a) Linear-wedge gray-scale image.
(b) Response of monitor to linear wedge.
(c) Gamma-corrected wedge.
(d) Output of monitor.



2 Intensity transformations

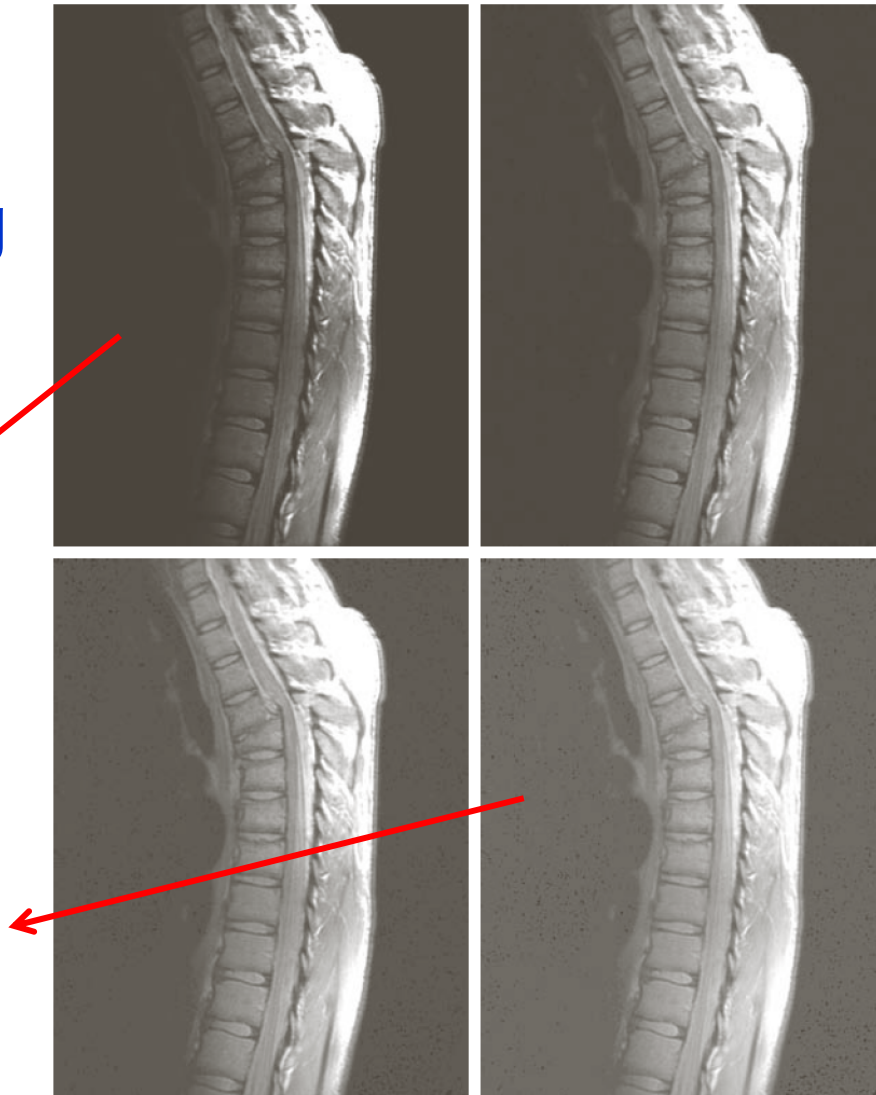
(3) Power-law (Gamma) Transformations

Contrast adjusting

Intensity levels
expansion needed

γ ?

Contrast decrease
Washed out!



a	b
c	d

FIGURE 3.8

(a) Magnetic resonance image (MRI) of a fractured human spine.
(b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 0.6, 0.4$, and 0.3 , respectively. (Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

Gamma Correction

Intensity levels
compression
needed

γ ?



a	b
c	d

FIGURE 3.9

(a) Aerial image.
(b)–(d) Results of
applying the
transformation in
Eq. (3.2-3) with
 $c = 1$ and
 $\gamma = 3.0, 4.0,$ and
 5.0 , respectively.
(Original image
for this example
courtesy of
NASA.)

Detail is
lost

2 Intensity transformations

(4) Piecewise-Linear Transformations

Complementary of several methods

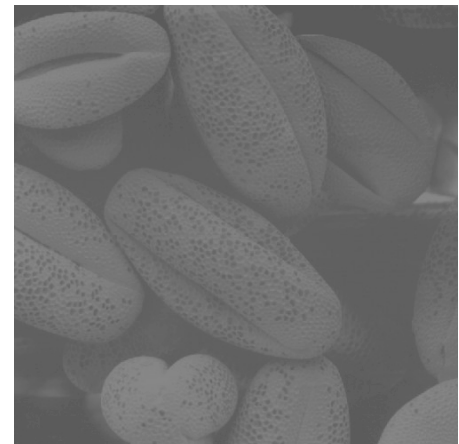
Contrast Stretching Transformation:

Expand the range of intensity levels to spans the full intensity range.

$$s = T(r) = mr + b$$

Low contrast image:

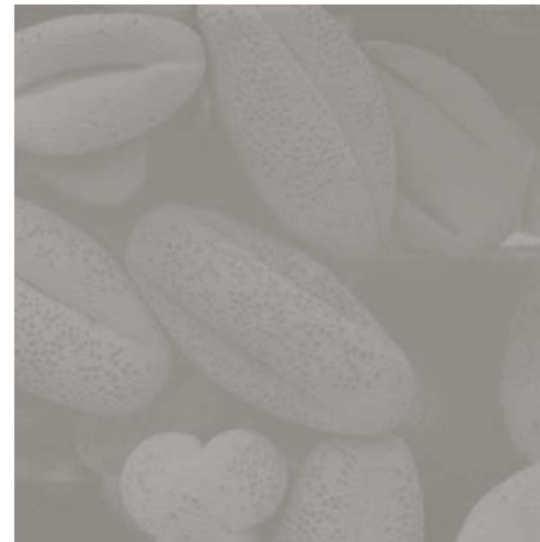
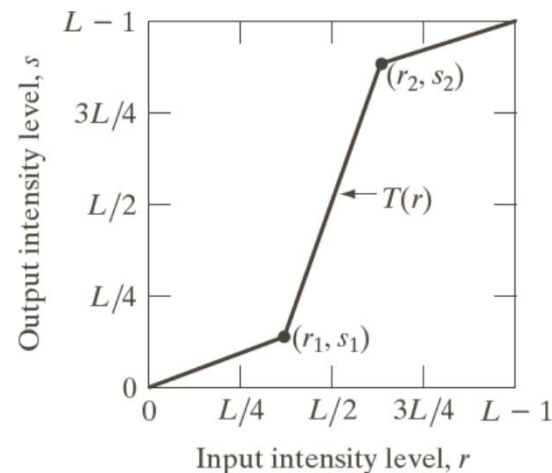
Poor illumination, lack of dynamic range in the image sensor and wrong setting of lens aperture.



2 Intensity transformations

(4) Piecewise-Linear Transformations

Location of points controls the shape of transformation of function.



$$\begin{aligned} Y_1 &= S_1, Y_2 = S_2 \\ &? \\ Y_1 &= Y_2, S_1 = 0 \\ S_2 &= L-1 ? \end{aligned}$$

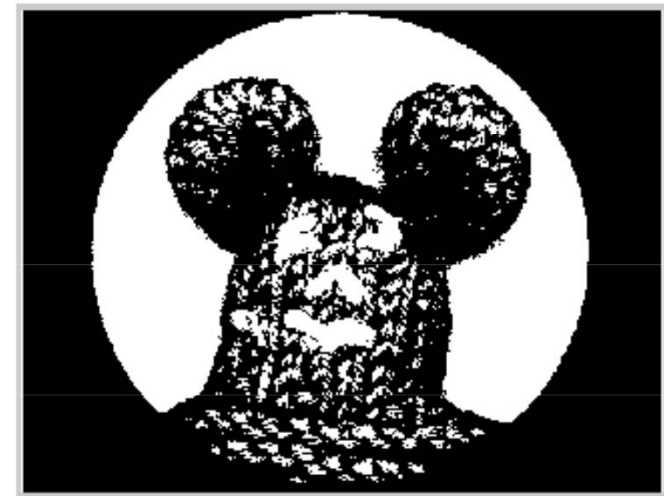
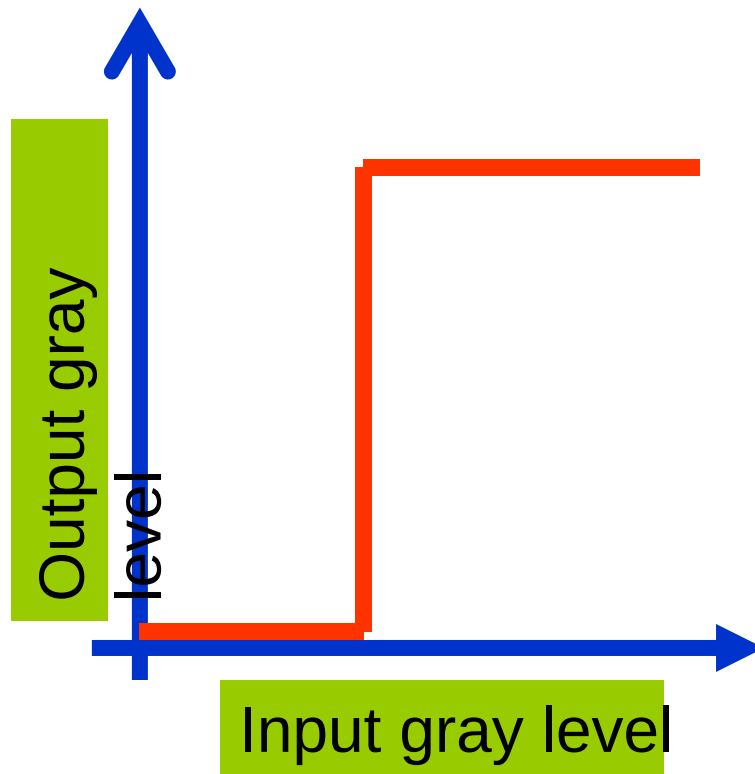
$$\begin{aligned} (r_1, s_1) &= (r_{\min}, 0), \\ (r_2, s_2) &= (r_{\max}, L-1) \end{aligned}$$



$$\begin{aligned} (r_1, s_1) &= (m, 0), \\ (r_2, s_2) &= (m, L-1) \end{aligned}$$

2 Intensity transformations

(4) Piecewise-Linear Transformations

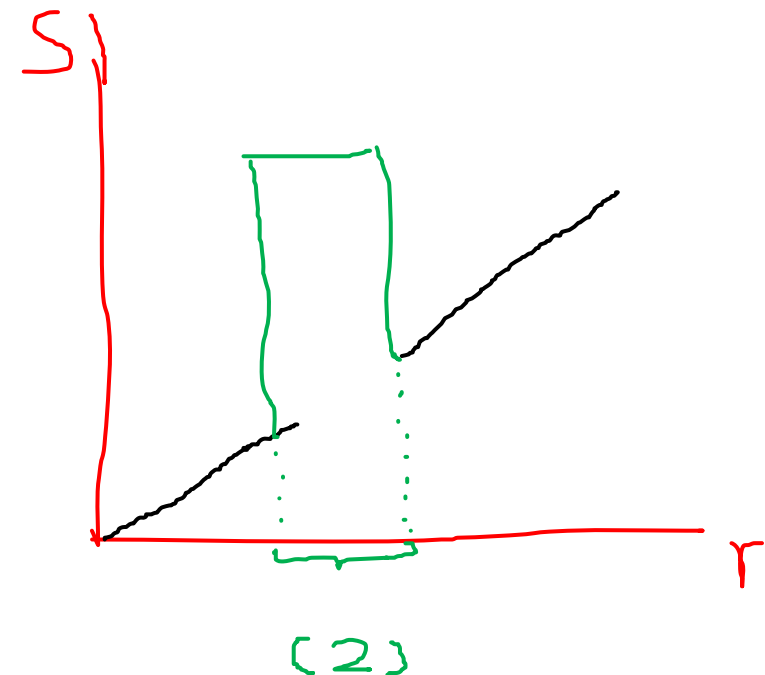
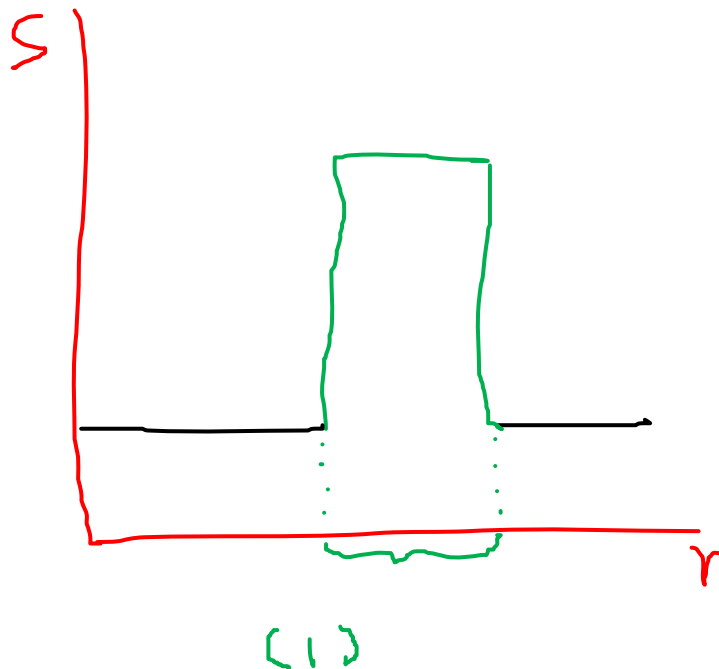


2 Intensity transformations

(4) Piecewise-Linear Transformations

Intensity-level slicing:

Enhancing certain features by: 1) one value all the values interested, and in another all other intensities. 2) brightens values interested, and leaves others unchanged.



2 Intensity transformations

(4) Piecewise-Linear Transformations

Intensity-level slicing:



a b c

FIGURE 3.12 (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. 3.11(a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. 3.11(b), with the selected area set to black, so that grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)

3 Image Histograms

3.1 Definition

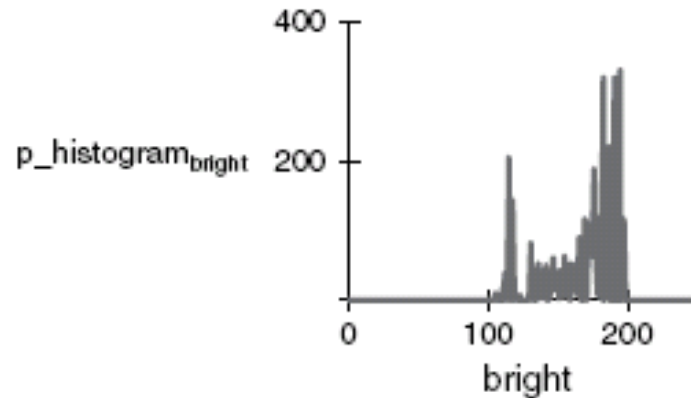
- A discrete function: **the distribution of grey levels in the image**
- Gray-level histogram: for each gray level, the number of pixels in the image have that gray level.

$$n_k = hist[r_k] = \sum_{f(x,y)=r_k} 1$$

- PDF: Normalized histogram (probability): $p_k = n_k / N$



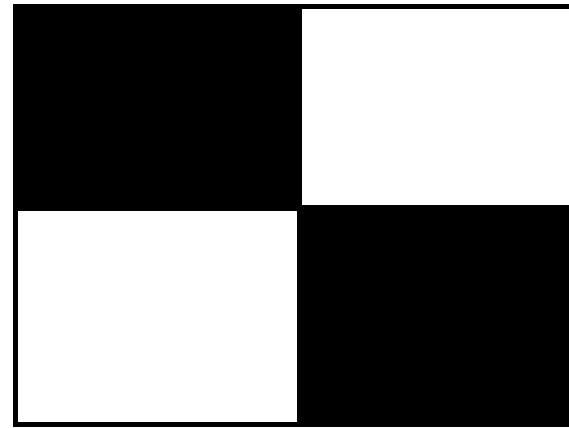
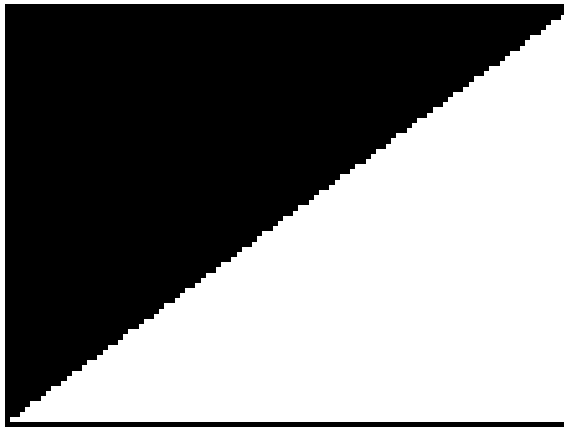
(a) Image of eye



(b) Histogram of eye image

x-axis – values of intensities

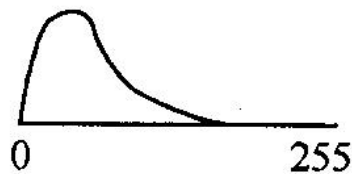
y-axis – their frequencies



- Q:
- Same histogram?
- Does histogram show the pixel location information in the image?

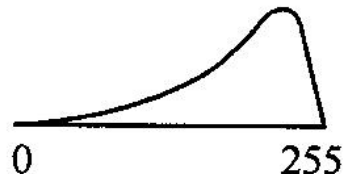
Histogram gives information about :

Bright or dark? Clear details? Dynamic range...



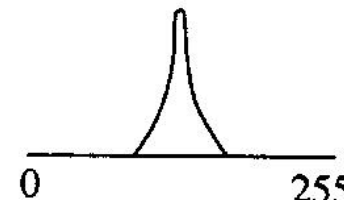
(a) 图象总体偏暗

Too dark



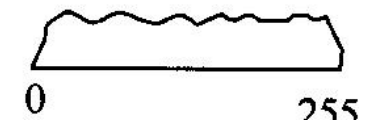
(b) 图象总体偏亮

Too bright



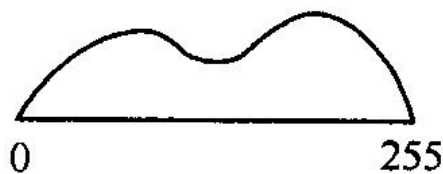
(c) 图象动态范围小，
细节不够清楚

Bad details



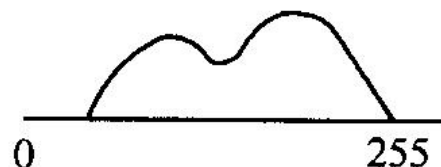
(d) 图象灰度分布均匀，
清晰明快

Clear details



(e) 图象动态范围适中

Medium dynamic range



(f) 图象动态范围偏小

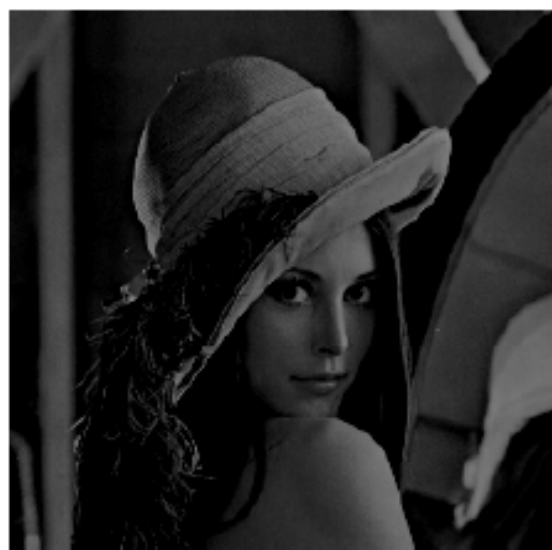
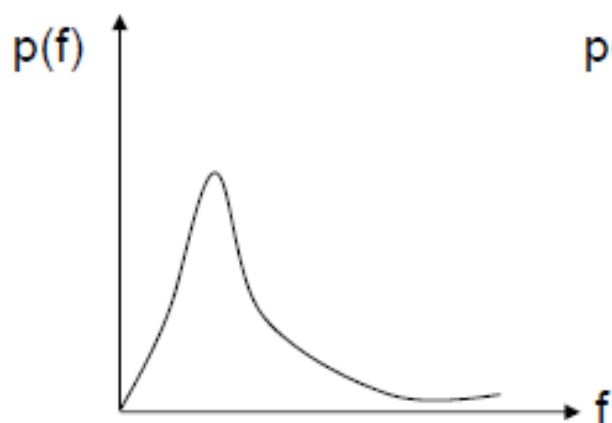
Small dynamic range



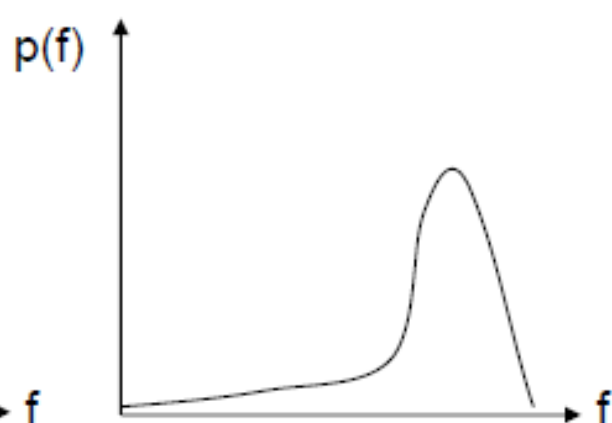
(g) 图象动态范围偏大

Wide dynamic range

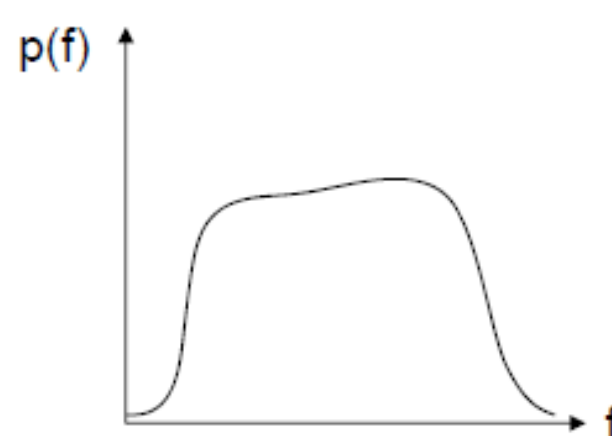
Examples of Histograms



(a) Too dark

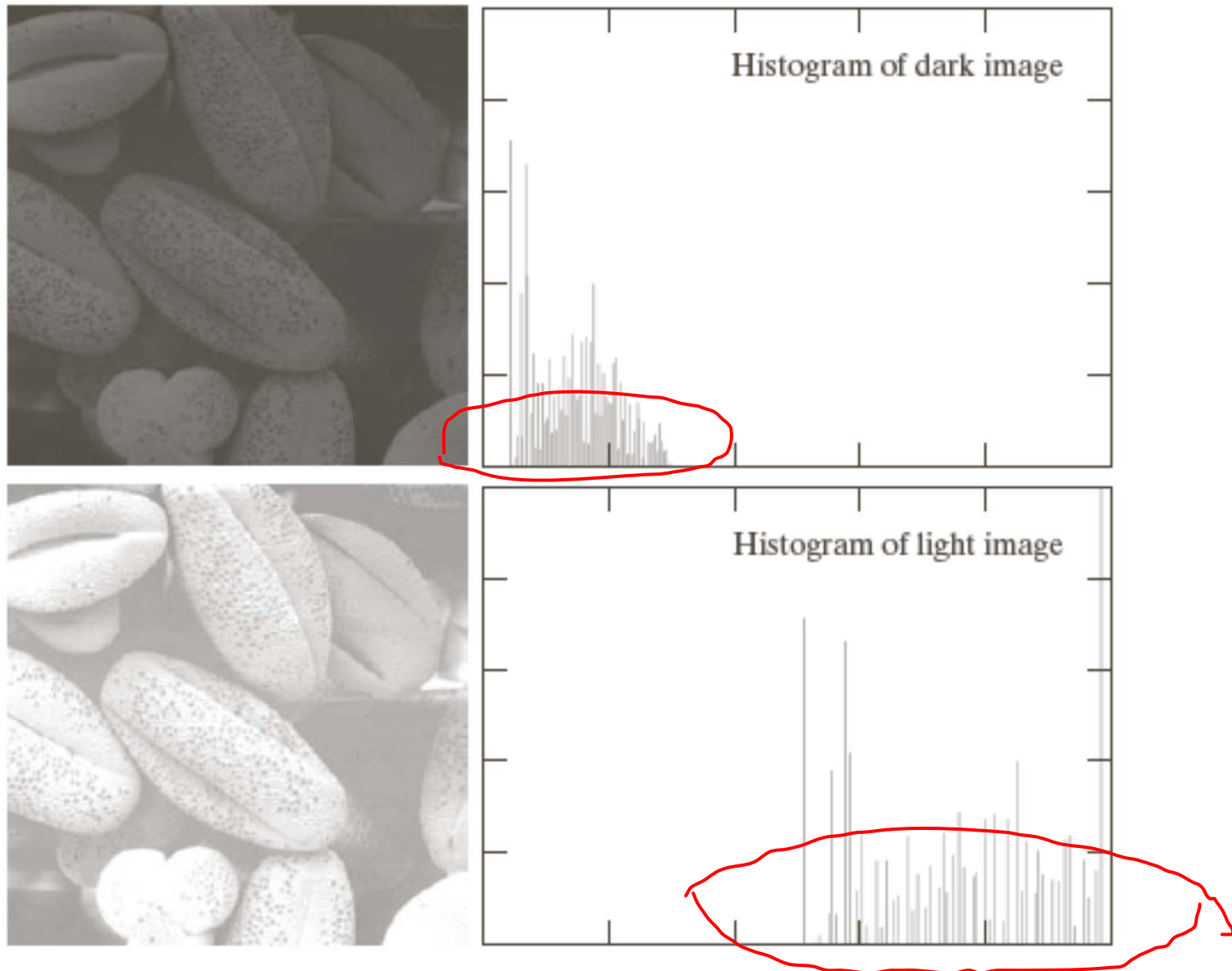


(b) Too bright

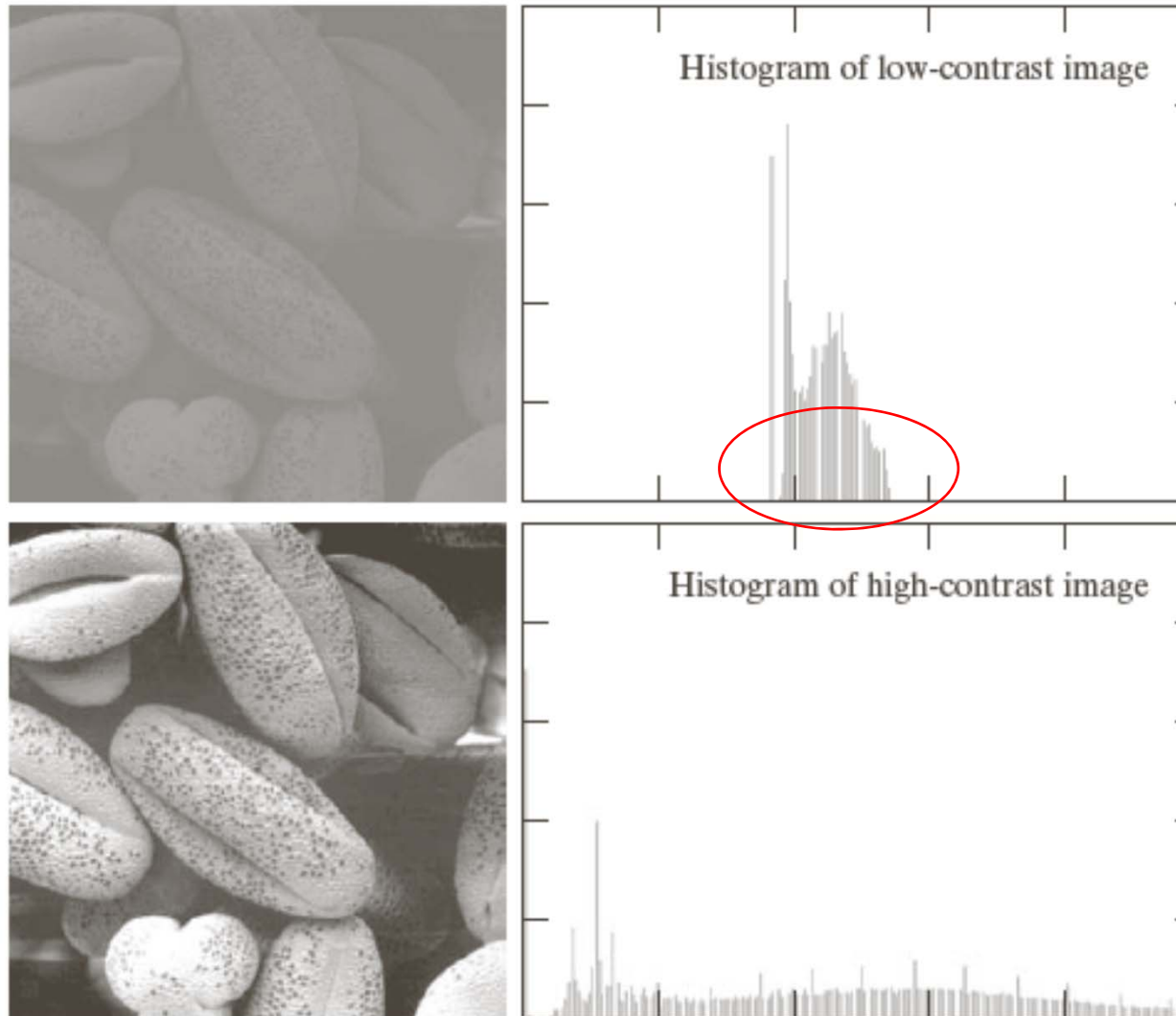


(c) Well balanced

histogram distribution



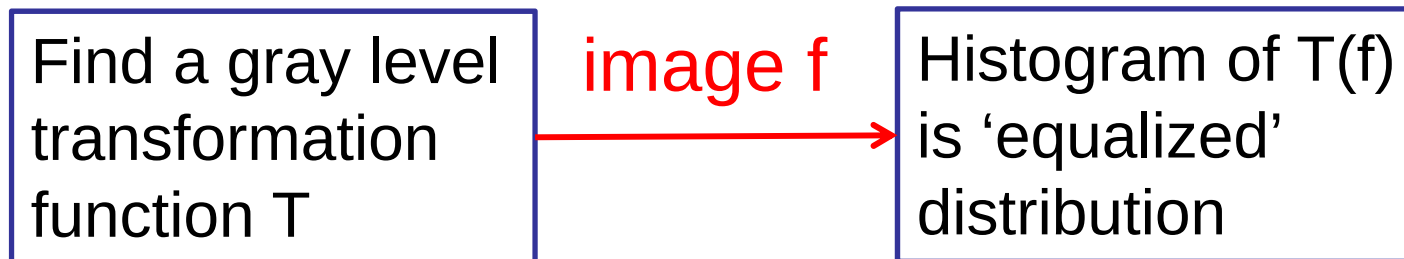
histogram distribution



Cover wide range of intensity

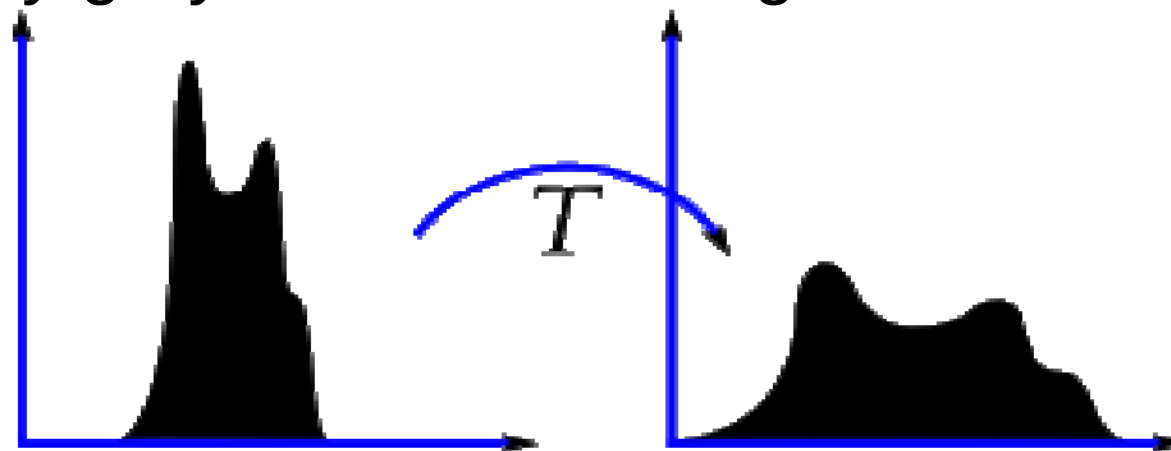
3.2 Histogram manipulation

What we want:



How?

The image consists of an equal number of pixels for every gray-value, the histogram is constant !



Histogram Equalization

To make the *PDF* of the transformed image uniform,
the formula for histogram equalization is given by

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k \frac{n_j}{n} = (L - 1) \sum_{j=0}^k p_r(r_j)$$

where

r_k : input intensity

s_k : processed intensity

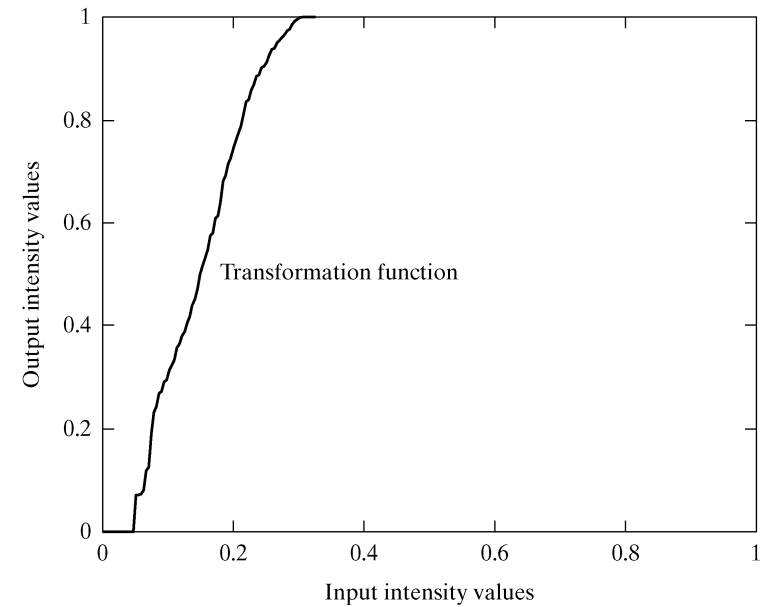
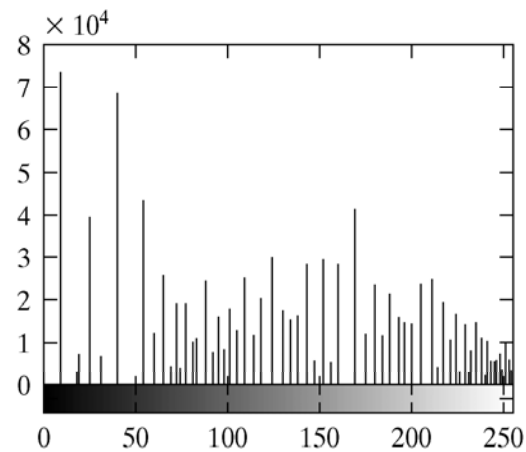
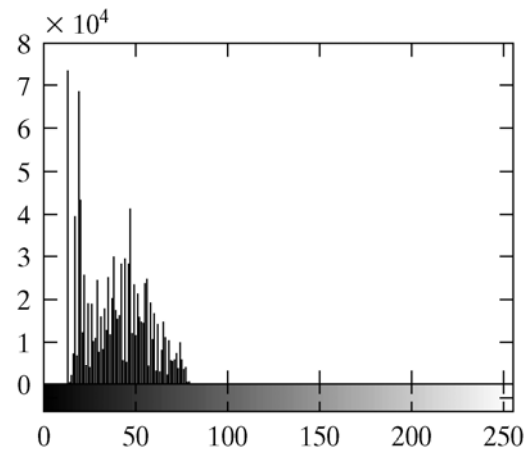
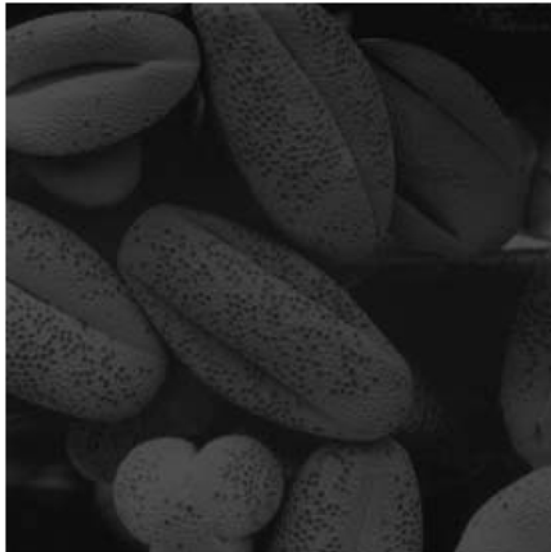
k : the intensity range (e.g 0.0 – 1.0)

n_j : the frequency of intensity j

n : the sum of all frequencies

$p_r(r_j)$: the **probability density function** (*PDF*)

$T(r_k)$: the **cumulative distribution function** (*CDF*)



Histogram Equalization example 1(from wiki)

52	55	61	66	70	61	64	73
63	59	55	90	109	85	69	72
62	59	68	113	144	104	66	73
63	58	71	122	154	106	70	69
67	61	68	104	126	88	68	70
79	65	60	70	77	68	58	75
85	71	64	59	55	61	65	83
87	79	69	68	65	76	78	94

Original Image



Histogram Equalization example 1(from wiki)

Histogram (PDF)

Value	Count	Value	Count	Value	Count	Value	Count	Value	Count
52	1	64	2	72	1	85	2	113	1
55	3	65	3	73	2	87	1	122	1
58	2	66	2	75	1	88	1	126	1
59	3	67	1	76	1	90	1	144	1
60	1	68	5	77	1	94	1	154	1
61	4	69	3	78	1	104	2		
62	1	70	4	79	2	106	1		
63	2	71	2	83	1	109	1		

Histogram Equalization example 1(from wiki)

cumulative distribution function (CDF)

Value	cdf	Value	cdf	Value	cdf	Value	cdf	Value	cdf
52	1	64	19	72	40	85	51	113	60
55	4	65	22	73	42	87	52	122	61
58	6	66	24	75	43	88	53	126	62
59	9	67	25	76	44	90	54	144	63
60	10	68	30	77	45	94	55	154	64
61	14	69	33	78	46	104	57		
62	15	70	37	79	48	106	58		
63	17	71	39	83	49	109	59		

CDF shows the minimum value in the sub-image is 52 and the maximum value is 154. The CDF of 64 for value 154 coincides with the number of pixels in the image.

$$h(v) = \text{round} \left(\frac{\text{cdf}(v) - \text{cdf}_{\min}}{(M \times N) - \text{cdf}_{\min}} \times (L - 1) \right)$$

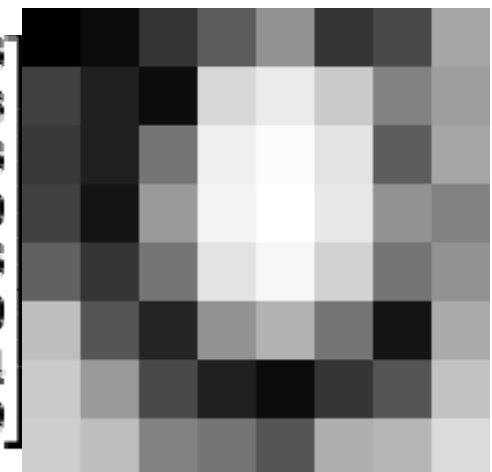
$$h(v) = \text{round} \left(\frac{\text{cdf}(v) - 1}{63} \times 255 \right)$$

For example, the cdf of 78 is 46, the normalized value becomes:

$$h(78) = \text{round} \left(\frac{46 - 1}{63} \times 255 \right) = \text{round}(0.714286 \times 255) = 182$$



0	12	53	93	146	53	73	166
65	32	12	215	235	202	130	158
57	32	117	239	251	227	93	166
65	20	154	243	255	231	146	130
97	53	117	227	247	210	117	146
190	85	36	146	178	117	20	170
202	154	73	32	12	53	85	194
206	190	130	117	85	174	182	219



Histogram Equalization Example 2

200	50	100
50	100	50

Input number pdf			cdf	Output
50	3	3/6	3/6=50%	255*50%=128
100	2	2/6	(3+2)/6=83.33%	255*83.33%=212
200	1	1/6	(3+2+1)/6=100%	255*100%=255

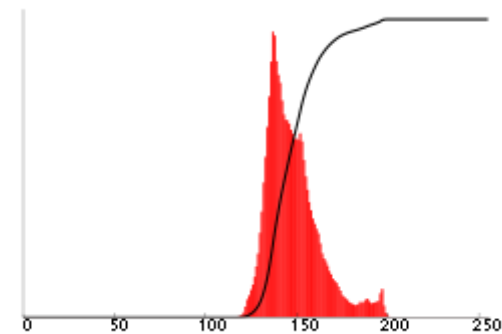
255	128	212
128	212	128

Histogram Equalization Example 3

An un-equalized image



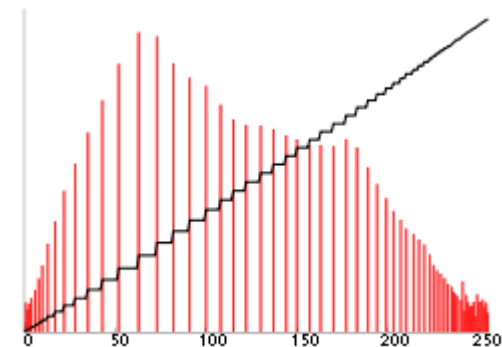
Corresponding histogram (red) and cumulative histogram (black)

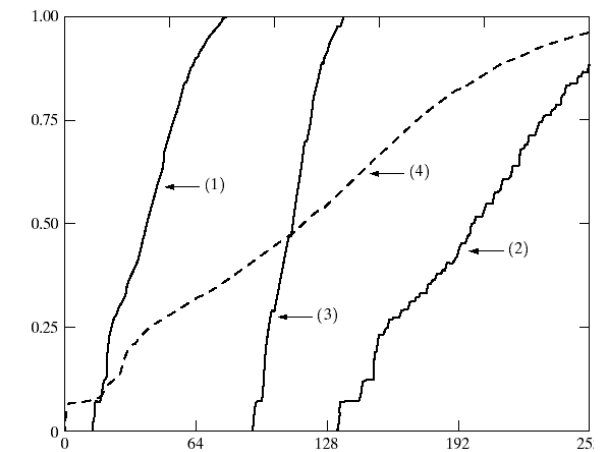
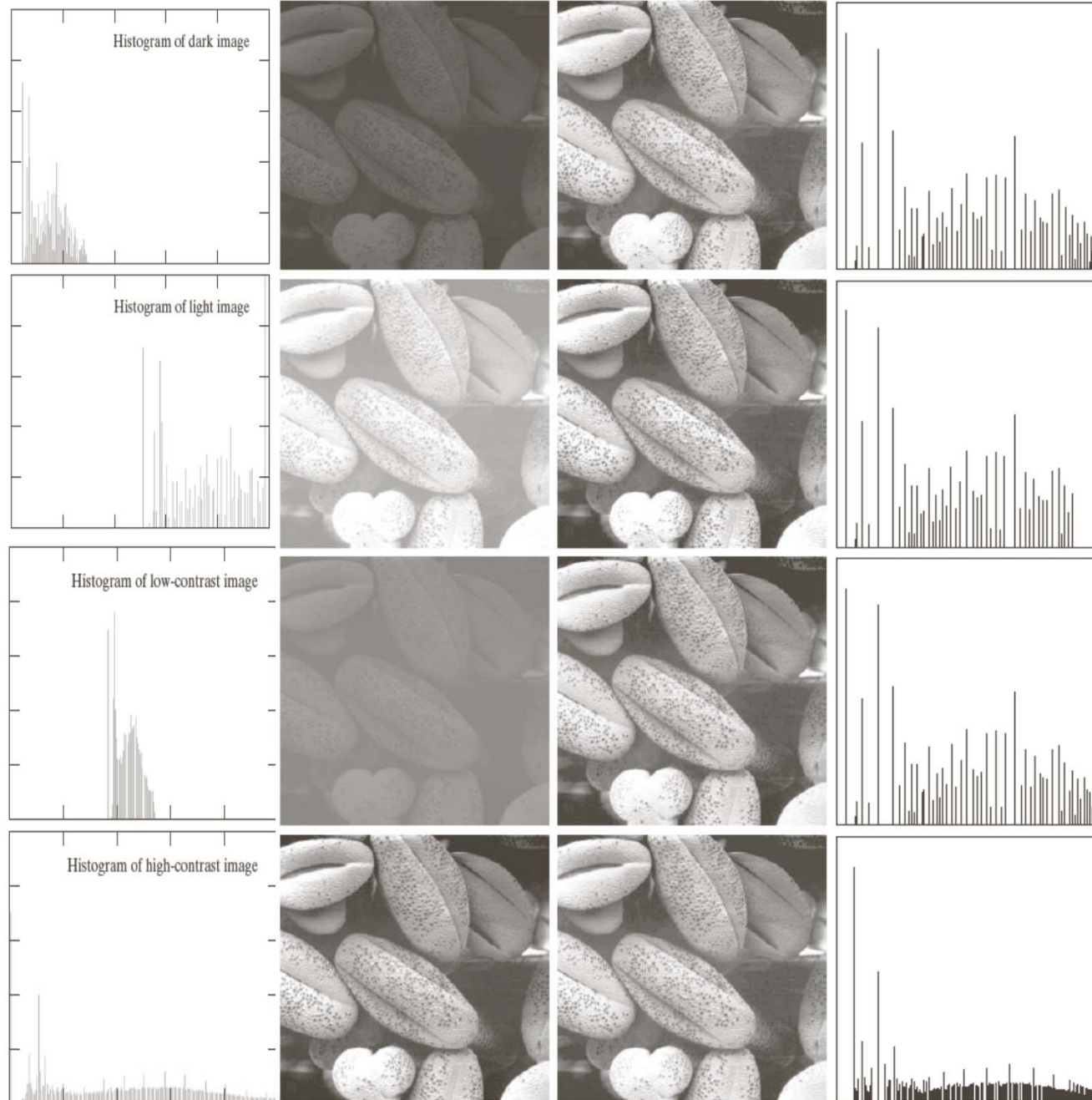


The same image after histogram equalization



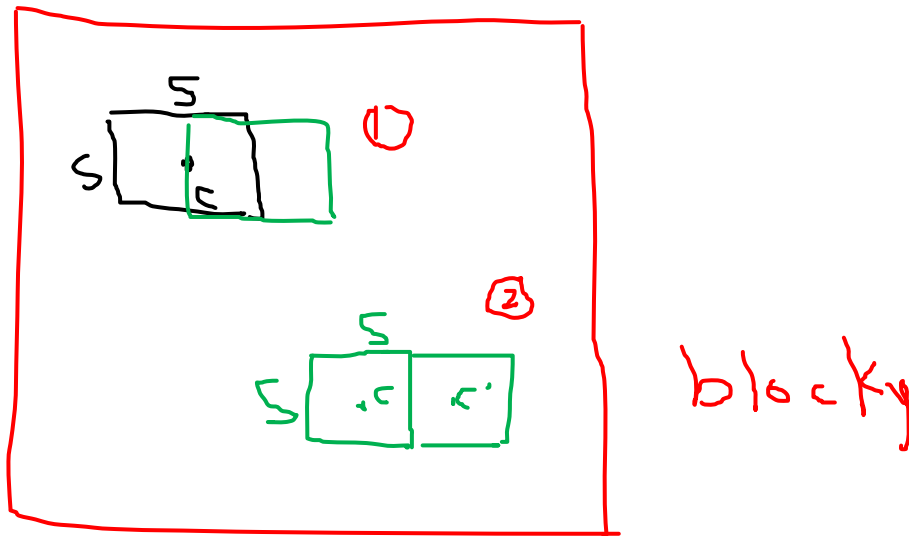
Corresponding histogram (red) and cumulative histogram (black)



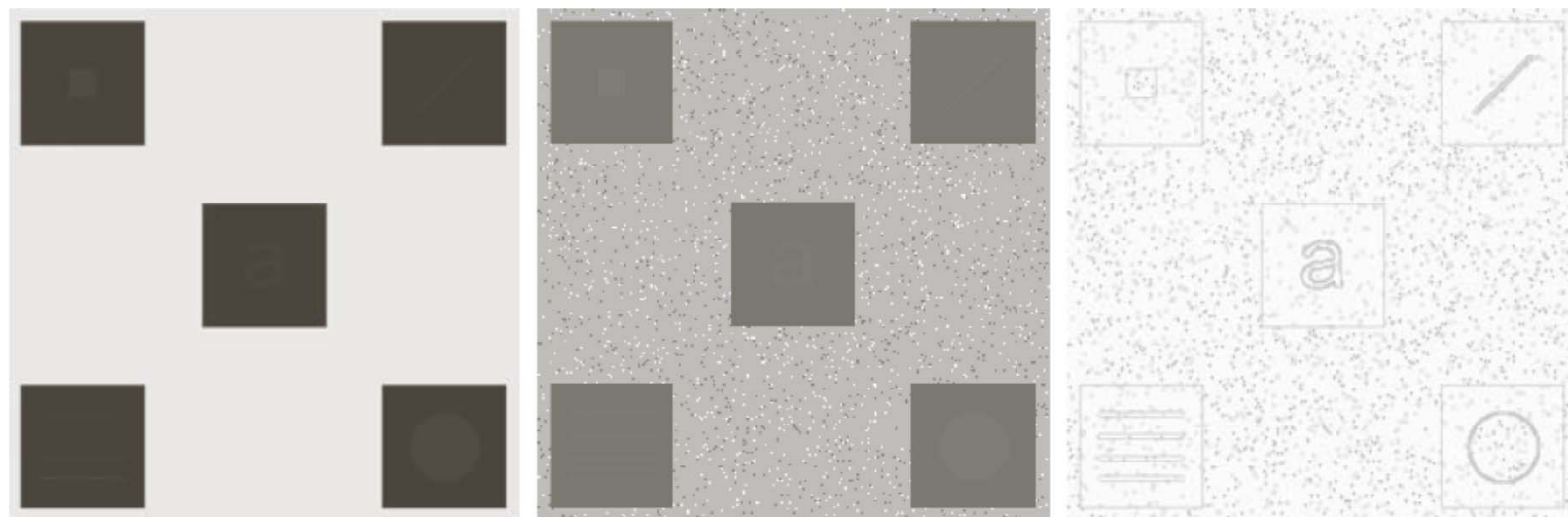


Local Enhancement:

- 1 Define square or rectangle around each pixel. Move center of window across the image.
- 2 At each location, compute histogram in the window. Then do histogram equalization.



Local Enhancement:



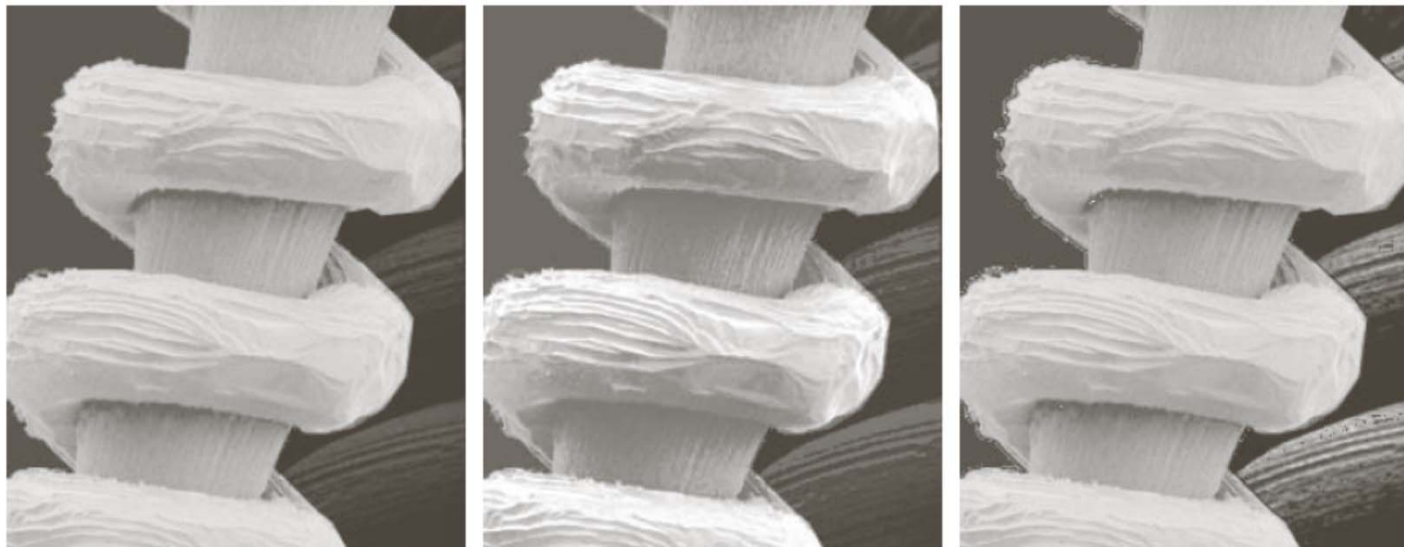
a b c

FIGURE 3.26 (a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization applied to (a), using a neighborhood of size 3×3 .

Histogram statistics for Image Enhancement:

Mean: average intensity

Variance: a measure of average intensity



a b c

FIGURE 3.27 (a) SEM image of a tungsten filament magnified approximately 130 $\times$. (b) Result of global histogram equalization. (c) Image enhanced using local histogram statistics. (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene.)

Problems:

1 When you enter a dark theater on a bright day, it take some time before you can see well enough to find a empty seat, explain the visual process.

2 Give four examples of energy source-image types pair according to different energy source.